

Q1.

- (a) Matthew is finding a formula for the inverse function $\operatorname{arsinh} x$.
He writes his steps as follows:

$$\text{Let } y = \sinh x$$

$$y = \frac{1}{2}(e^x - e^{-x})$$

$$2y = e^x - e^{-x}$$

$$0 = e^x - 2y - e^{-x}$$

$$0 = (e^x)^2 - 2ye^x - 1$$

$$0 = (e^x - y)^2 - y^2 - 1$$

$$y^2 + 1 = (e^x - y)^2$$

$$\pm\sqrt{y^2 + 1} = e^x - y$$

$$y \pm \sqrt{y^2 + 1} = e^x$$

To find the inverse function, swap x and y : $x \pm \sqrt{x^2 + 1} = e^y$

$$\ln(x \pm \sqrt{x^2 + 1}) = y$$

$$\operatorname{arsinh} x = \ln(x \pm \sqrt{x^2 + 1})$$

Identify, and explain, the error in Matthew's proof.

- (b) Solve $\ln(x + \sqrt{x^2 + 1}) = 3$

(2)

(1)

(Total 3 marks)

Q2.

- (a) Using the definition $\tanh y = \frac{e^y - e^{-y}}{e^y + e^{-y}}$, show that, for $|x| < 1$,

$$\tanh^{-1} x = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right)$$

(3)

- (b) Hence, or otherwise, show that $\frac{d}{dx} (\tanh^{-1} x) = \frac{1}{1-x^2}$.

(3)

- (c) Use integration by parts to show that

$$\int_0^{\frac{1}{2}} 4 \tanh^{-1} x \, dx = \ln \left(\frac{3^m}{2^n} \right)$$

where m and n are positive integers.

(5)

(Total 11 marks)

Q3.

- (a) (i) Show that

$$\frac{d}{du} (2u\sqrt{1+4u^2} + \sinh^{-1} 2u) = k\sqrt{1+4u^2}$$

where k is an integer.

(4)

- (ii) Hence show that

$$\int_0^1 \sqrt{1+4u^2} \, du = p\sqrt{5} + q \sinh^{-1} 2$$

where p and q are rational numbers.

(2)

- (b) The arc of the curve with equation $y = \frac{1}{2} \cos 4x$ between the points where $x = 0$ and $x = \frac{\pi}{8}$ is rotated through 2π radians about the x -axis.

- (i) Show that the area S of the curved surface formed is given by

$$S = \pi \int_0^{\frac{\pi}{8}} \cos 4x \sqrt{1 + 4 \sin^2 4x} \, dx$$

(2)

- (ii) Use the substitution $u = \sin 4x$ to find the exact value of S .

(4)

(Total 12 marks)

Q4.

- (a) Sketch the curve $y = \cosh x$.

(1)

- (b) Solve the equation

$$6 \cosh^2 x - 7 \cosh x - 5 = 0$$

giving your answers in logarithmic form.

(6)

(Total 7 marks)

Q5.

- (a) Use the definitions of $\cosh \theta$ and $\sinh \theta$ in terms of e^θ to show that

$$\cosh x \cosh y - \sinh x \sinh y = \cosh(x - y)$$

(4)

- (b) It is given that x satisfies the equation

$$\cosh(x - \ln 2) = \sinh x$$

- (i) Show that $\tanh x = \frac{5}{7}$.

(4)

- (ii) Express x in the form $\frac{1}{2} \ln a$.

(2)

(Total 10 marks)

Q6.

- (a) Use the definitions $\cosh x = \frac{1}{2}(e^x + e^{-x})$ and $\sinh x = \frac{1}{2}(e^x - e^{-x})$ to show that

$$\cosh^2 x - \sinh^2 x = 1$$

(3)

- (b) (i) Express

$$5 \cosh^2 x + 3 \sinh^2 x$$

in terms of $\cosh x$.

(1)

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- (ii) Sketch the curve $y = \cosh x$.

(1)

- (iii) Hence solve the equation

$$5 \cosh^2 x + 3 \sinh^2 x = 9.5$$

giving your answers in logarithmic form.

(4)

(Total 9 marks)

Q7.

- (a) Show that

$$\frac{d}{dx} \left(\cosh^{-1} \frac{1}{x} \right) = \frac{-1}{x\sqrt{1-x^2}}$$

(3)

- (b) A curve has equation

$$y = \sqrt{1-x^2} - \cosh^{-1} \frac{1}{x} \quad (0 < x < 1)$$

Show that:

(i) $\frac{dy}{dx} = \frac{\sqrt{1-x^2}}{x}$;

(4)

- (ii) the length of the arc of the curve from the point where $x = \frac{1}{4}$ to the point where $x = \frac{3}{4}$ is $\ln 3$.

(5)

(Total 12 marks)

Q8.

- (a) Express

$$5 \sinh x + \cosh x$$

in the form $Ae^x + Be^{-x}$, where A and B are integers.

(2)

- (b) Solve the equation

$$5 \sinh x + \cosh x + 5 = 0$$

giving your answer in the form $\ln a$, where a is a rational number.

(4)

(Total 6 marks)